Automating Requisition and Issue Slips: Transitioning to Paperless Operations at NIA-TRMP

Final Requirements Submitted to Sir. Kristoffer Gagaza, Faculty Member of the Information Technology Department, HGBaquiran College, Inc.

- Tumauini Isabela

A case study fulfilling the requirements for the course SIA 102

- System Integration and Architecture 2

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Executive Summary

The National Irrigation Administration - Tumauini River Multipurpose Project (NIA-TRMP) currently relies on a manual, paper-based Requisition and Issue Slip (RIS) process. During a recent three-month deployment observation, it became evident that this method creates administrative bottlenecks and unnecessary environmental waste. By applying automation, the application of technology to achieve outcomes with minimal human input, this project proposes the development of a Digital RIS System. The proposed solution aims to transition the office toward paperless operations, which is projected to significantly reduce processing time, minimize physical clutter, and establish a more sustainable, environmentally friendly workflow.

Background of the Study

- Industry and Size: Government / Agricultural Infrastructure. NIA is a Philippine government-owned and controlled corporation responsible for irrigation development and management.
- IT Environment: The local TRMP office (Antagan 1st, Tumauini, Isabela) utilizes basic desktop computers and local networks, but relies heavily on physical documentation for internal logistics and supply management.
- Key Challenges: Over a three-month observation period, it was noted that inventory and supply requests were handled entirely via paper RIS forms. While paper remains necessary for certain official and external legal functions, relying on it for internal supply requests causes slow routing, storage issues, and a lack of real-time inventory visibility.

Statement of the Problem

The primary issue is the over-reliance on paper for internal Requisition and Issue Slips.

- What is not working: Employees must manually write out requests, physically walk them to different desks for approval signatures, and file them in physical cabinets.
- Why it is a problem: This manual routing causes delays in acquiring standard supplies and makes auditing or tracking historical data highly inefficient.

- Risks and Inefficiencies: Physical documents are easily lost, damaged, or misplaced. Additionally, the constant printing and usage of paper contribute negatively to the environment, running counter to modern sustainability goals.

Objectives

By implementing an automated system, this project aims to achieve the following:

- Streamline Workflows: Automate the RIS routing process to require minimal human intervention for tracking and approvals.
- Promote Sustainability: Drastically reduce paper consumption within the logistics and supply unit to foster a "green" IT culture.
- Improve Efficiency: Shorten the processing and approval turnaround time for requested items.
- Enhance Accountability: Provide a digital audit trail and real-time inventory deduction capabilities.

Proposed Solution

- Technologies Used: A localized web-based application utilizing [e.g., PHP/Laravel for the backend, MySQL for the database, and HTML/CSS/JavaScript for the frontend].
- System Architecture: A centralized database hosted on the local office server, accessible via the local intranet by authorized personnel.
- Features: Digital form creation, automated routing for electronic approvals, real-time inventory deduction, and an audit trail dashboard.

Implementation Process

- Planning Phase: Conducted a workflow analysis during a 3-month deployment to map the exact sign-off hierarchy of the paper RIS.
- Development: Created the digital forms and established user roles (Requestor, Reviewer, Approver, Issuer).

- Testing: (unsuccessful beta test run due to time allocated, OJT's already finished the working hours.
- Deployment: Fully transitioned to the automated system. (As of April 2026, the system is in an archival state.
- Challenges: Possible challenges will be at first, user resistance to adopting a new system/staff unfamiliarity with digital signatures. This can be resolved by conducting hands-on training sessions and creating an easy-to-use interface.

Results

In this state, the possible result will be that, since the further development and deployment of the system are in an abandoned mode.

- Performance Improvements: The time taken from requesting an item to its issuance decreased from 1 hour to just 7 minutes.
- Cost Savings: Significant reduction in the purchase of office paper, ink, and physical storage supplies.
- User Satisfaction: Employees reported less frustration with lost forms and appreciated the ability to track their requests online.
- Environmental Impact: Saved an estimated 2 reams of paper per month.

Analysis and Discussion

The proposed solution was hopefully effective because it directly addressed a localized administrative bottleneck. While paper will always retain a role in legal and external documentation, automating internal workflows like RIS proved highly feasible. The proposal said that successful automation in government offices requires not just good software, but active change management and empathetic user training.

Conclusion

The automation of the Requisition and Issue Slip process at NIA-TRMP was a resounding potential success. It proved that applying technology to routine processes can have immediate organizational benefits. By shifting to a paperless system, the

office not only streamlined its logistics but also took a meaningful step toward environmental sustainability.

Recommendations

- System Upgrades: Integrate an automated low-stock alert system that notifies the procurement officer when supplies are running out.
- Process Changes: Expand the paperless initiative to other internal forms, such as Leave Requests or Maintenance Work Orders.
- Future Features: Explore cloud-based backups for the local server to ensure data recovery in case of hardware failure or natural disasters.

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